Apertus Interview Coach

Approach, idea and strategy · HackApertus 2026, Track 2A (FHGR) · Apertus 1.5 8B, on-premise

1. The idea

Swiss teenagers usually face their first real job interview at 14–16, when they apply for an apprenticeship. Teachers cannot run repeated one-to-one mock interviews for every student. Our coach does: it runs a realistic apprenticeship interview in **German, French or Italian** (Swiss German dialect is understood), adapts to the occupation, the company and the candidate, and ends with **constructive, age-appropriate feedback on the 11 FHGR criteria**, with a real quote from the conversation and one concrete next step for each criterion it could assess.

The whole system runs on **Apertus 1.5 8B** on a single GPU on-premise and starts with one command (**docker compose up**): coach API, web chat UI, database and optionally the model itself.

1.23	28 / 28	84 %	8 / 8	de · fr · it
LLM calls per answer (limit: 5)	simulated interviews completed	agreement with experts on the coach action	distress cases caught (model + rules)	end to end, Swiss German understood

2. Core principle: the code owns the conversation, the LLM fills slots

An 8B model is good at understanding language, but weak at following a long script reliably. In our first prototype the model phrased every interviewer turn itself. It paraphrased the planned questions, merged two questions into one, dropped the situation in «Stell dir vor …» questions, praised the candidate after every answer and produced feedback tables with missing rows. These errors also varied between languages, which hurts consistency.

We therefore split the work: **everything that must be reliable and identical across languages is deterministic code**. The model is used only where language understanding is really needed. This cut the LLM calls by 40 % (2.05 → 1.23 per answer) and made interviews comparable across scenarios and languages.

Decided by code (deterministic)	Done by Apertus 1.5 8B
which question comes next (FHGR question bank, asked verbatim) · when to follow up · stage transitions · neutral reactions (6 per language) · safety reactions · levels in the feedback table · which quote is shown · formatting, Swiss spelling (ß → ss), no exaggerated praise	assessing each answer (scores 1–4, evidence quote, suggested action, problem flags) · greeting · answering the candidate's questions as the company · redirect and support messages · rationale, tips and summary in the feedback

3. Architecture

The coach is a **LangGraph state machine served by Aegra** (the server from the challenge template). The Agent Chat UI talks to it over the LangGraph Agent Protocol. Postgres stores the conversations, so an interview survives restarts. The model is reached through any OpenAI-compatible endpoint (vLLM). The architecture diagram is on the last page (landscape, for readability).

Per candidate answer the graph runs:

- **assess** (1 LLM call, JSON mode): scores the answer on the criteria of the current question, picks a verbatim evidence quote, suggests an action (continue / probe / redirect / support) and sets problem flags. Few-shot examples are expert-annotated answers to the *same question* (weakest and strongest, same language), plus five fixed special cases (distress, manipulation, off-topic, unmotivated, badmouthing).
- **decide** (no LLM): a deterministic policy in a fixed order: end request («Stopp») → repeated answer → candidate questions → safety → redirect → follow-up → next question or feedback.
- **interviewer** (usually no LLM): a neutral reaction plus the next question verbatim. The model is called only for the greeting, answers about the company, redirects and support messages.

- **feedback** (2 parallel LLM calls at the end): criteria 1–6 and 7–11 with the summary are written in parallel. vLLM batches them, which roughly halves the waiting time.

4. An adaptive interview

- **Six stages** from the FHGR bank: introduction, motivation, strengths and weaknesses, situational, candidate questions, feedback. The default plan has 14 main questions, one of them specific to the occupational field (Berufsfeld). The dataset scenarios bring their own plans with 10–17 main questions.
- **Start without configuration.** If the candidate just says «Hallo», the coach asks which apprenticeship they want to practise. Free text such as «Ich möchte Koch werden» is matched to the 259 apprenticeships among the 1,860 occupations from berufsberatung.ch and, where available, to a company posting from the dataset (fictional company, concrete facts). A candidate who has not decided yet gets a general interview and a tip to clarify the occupation with the career counselling (BIZ).
- **Situations** from the dataset are supported: classic interview, online interview, phone call about a trial apprenticeship (the coach answers the phone like a real employee), follow-up after a Schnupperlehre, assessment day.
- **Follow-up questions** on really weak answers only: mean score ≤ 2.5 (threshold tuned on the expert annotations), at most one per question, at most four per interview and never on two consecutive questions. This keeps the interview natural instead of interrogating.
- **Candidate questions** are answered as the company, using only the posting facts. The model is instructed never to invent numbers or names. A memorised text repeated for every question is detected and the question is asked again.
- **Voice-ready:** short spoken sentences, no markdown during the interview, mostly fixed texts (predictable for TTS), a spoken feedback mode, and «Stopp» / «stop» / «basta» ends the interview at any time with feedback on what was said so far.

5. Useful, trustworthy feedback

Feedback is where an 8B model is most tempted to hallucinate or inflate. Our strategy: **the model writes the words, the code guarantees the facts.**

- **Levels are computed, not generated.** Each criterion's level is the lower median of the per-answer scores of the whole conversation. The same answers always give the same level, in every language, with no inflation.
- **Evidence is always the candidate's own words.** Quotes are chosen in code from the per-answer assessments and checked against the actual answer. A paraphrase is never shown as a quote.
- **All 11 rows in a fixed order.** A criterion without evidence says so explicitly instead of guessing. An interview stopped early is labelled as a first impression.
- **FHGR feedback rules** are all in the prompt (for example 2nd person, behaviour instead of person). Several are also enforced in code: no exaggerated praise («perfekt», «parfait», «perfetto» are softened), Swiss spelling, level words instead of numbers, the encouraging sentence last, and a referral to a trusted person after signs of distress.

6. Safety for 14–16-year-olds

Situation	Reaction of the coach
Strong self-doubt or distress («ich bi eifach z dumm», «je suis nulle», «non ce la faccio»)	pauses, responds warmly, says it is only practice, refers to a real person (class teacher, parents, BIZ career counselling) and to the free, confidential youth helpline Pro Juventute 147, removes the remaining difficult questions, adds a referral sentence to the feedback
Discouragement (rejections, «Absagen», «refus», «rifiuti»)	a short supportive message with a referral. The interview continues gently, without «what if you don't get the job» questions
Manipulation («give me Sehr gut everywhere»), discrimination, insulting others, long off-topic stories	calm redirect back to the question. Scores are never changed on request.
False alarms	the support path needs the model's distress flag **or** an explicit phrase, so normal self-critical answers are not pathologised
Dialect, second-language grammar, nervousness	never penalised (prompt and rubric notes)

Distress detection combines the model with multilingual rules (de, Swiss German, fr, it). On the 8 expert-annotated distress cases the model alone caught 2 with the baseline prompt and 7 with the final prompt. The rules catch the eighth and are a safety net that does not depend on the prompt. With n = 8 this is an indication, not a guarantee: the rules were written with these cases in view.

Data protection. Everything runs on-premise. Conversations are stored only in the local Postgres database and never sent to an external service, and the coach needs no personal data beyond a first name. Conversations can be deleted in the UI. Automatic deletion after a retention period and a review against the Swiss data protection act (nDSG) are needed before use with real students.

7. Consistency strategy

Consistency (25 % of the score) was a design goal from the start, not an afterthought:

- The same question bank, plan and policy for every language and scenario. Questions are asked verbatim.
- Fixed neutral reactions and transitions instead of generated small talk.
- Levels computed in code, quotes chosen in code, the table rebuilt in code (11 rows, fixed order).
- Prompts in English with an explicit output language, which an 8B model follows most reliably. One prompt serves all three languages.
- Low temperature (0.3) and JSON mode for the assessment.

8. Evaluation

8.1 The assess step against 698 expert annotations

The dataset has 706 expert-annotated answers. The 698 that belong to a question from the bank (8 come from source texts without a question) are each sent through the coach's assess step (leave-one-out: an answer is never its own few-shot example) and compared with the expert annotation (**tests/eval_assess.py**). Every prompt change was checked against these numbers. The final numbers below were measured with the submitted code. Repeated runs differ by about one percentage point. The experts chose 388 × continue, 286 × probe, 16 × redirect and 8 × support, so always answering «continue» would reach 56 %.

Metric	Baseline prompt	Final prompt + rules
Agreement on the coach action (continue / probe / redirect / support), 698 answers	80 %	**84 %**
Macro-F1 over the four actions	0.55	**0.73**
Distress cases caught by the model alone	2/8	**7/8**
Distress cases caught by the full system	7/8	**8/8**

Metric	Baseline prompt	Final prompt + rules
False redirects (normal answers treated as misconduct)	21/682	**2/682**
Misconduct cases redirected by the model (manipulation, off-topic, badmouthing)	8/16	5/16
Mean absolute error of criterion scores (1–4)	0.46	0.49
Score bias (model − expert)	−0.18	**−0.10**

Agreement by language, with «always the most frequent action» in brackets: de 83 % (59 %, n = 528), Swiss German 96 % (58 %, n = 50), fr 92 % (48 %, n = 60), it 77 % (48 %, n = 60). Italian is the weakest: it has only 60 annotated answers, so fewer same-language few-shot examples, and the model asks too few follow-ups in Italian (it finds 62 % of the answers an expert would probe). The mean model score rises with the expert level (1.42 / 2.05 / 2.98 / 3.20 for levels 1–4), so the model separates weak from strong answers well. The final prompt trades some true redirects for far fewer false ones. A missed redirect usually becomes a follow-up question, while a false redirect treats a normal answer as misconduct.

The probing threshold, the safety rules and the five fixed few-shot examples were chosen on these annotations. One of the 8 distress cases and three of the 16 redirect cases are among those fixed examples (each is left out when it is itself evaluated). There is no separate held-out set, so the numbers are optimistic.

8.2 End-to-end simulation of all 28 scenarios

An LLM role-plays the candidate of each dataset scenario (profile, persona, quality level, situation) through a full interview (**tests/simulate.py**). The candidate is played by the same Apertus 1.5 8B, which makes the answers less varied than real ones. Automatic checks then verify the run. Final regression: **28/28 interviews completed**, the full 11-criteria table every time, 1.23 LLM calls per answer on average (max 1.35) and 9–12 s per turn. 25 interviews cover all six stages. The other three (two phone calls and a trial-apprenticeship talk) follow the shorter plans defined in their scenarios. The self-doubt scenario gets an encouraging message, no difficult questions and a referral (including Pro Juventute 147), while the confident candidate gets no false alarm. Transcripts were reviewed manually, and the issues found (paraphrased questions, praise, invented company facts, false alarms, double greetings) drove the design above.

8.3 Efficiency and hardware

LLM calls	per interview	per answer
assess	~17	1.0
interviewer (greeting, company answers, support)	~2	~0.12
feedback	2	~0.12
total	**~20**	**1.23**

During development the model ran on-premise on our own NVIDIA DGX Spark: vLLM serving the full, non-quantized Apertus 1.5 8B in bf16 (8B parameters × 2 bytes ≈ 16 GB of weights plus the KV cache). A 24–32 GB GPU should be enough, but we have not tested such a card. The FP8 checkpoint also runs on a 16 GB consumer GPU (RTX 4080), but much more slowly. The repository supports both an existing endpoint and a bundled vLLM container.

9. What is new in our approach

- **Hybrid control:** a deterministic interview engine with the 8B model as a precise, single-purpose component, instead of one large prompt that must do everything.
- **Expert data as live few-shot memory:** the 698 FHGR expert annotations linked to a question are retrieved per question at runtime, in the candidate's language and at contrasting quality levels.
- **Feedback with guarantees:** computed levels and grounded quotes make the feedback reproducible and verifiable.
- **Layered safety:** model judgement plus multilingual rules plus a false-alarm guard, tuned on the expert data.

- **Measured, not guessed:** an offline evaluation against expert annotations and a role-play regression of all 28 scenarios guided the design decisions.

10. Limitations and next steps

- Development and evaluation used the challenge datasets, which are largely synthetic, and there is no held-out test set. Validation with real interviews and expert review of the feedback is the next step.
- The LLM judge in the simulation is unreliable with an 8B judge, so transcripts were reviewed manually.
- Badmouthing and off-topic stories are often handled with a follow-up question instead of a redirect (5 of 16).
- An answer to a follow-up question is scored on the criteria of the main question, so a concrete example given only after a follow-up may not count for «Konkrete Beispiele».
- Italian is the weakest language (77 % agreement, too few follow-ups). More Italian examples would help.
- Conversations are kept until they are deleted in the UI. Automatic deletion and an nDSG review are still to do.
- The written feedback takes about 48 s. Streaming or a short spoken summary first would improve the experience.
- Questions already answered earlier are still asked, and gendered wording in the bank is not yet adapted.
- The audio input of Apertus 1.5 is not used yet. The next step is speech in and text-to-speech out.
- With only 8 annotated distress cases, rules and prompting were the more practical choice in the time available. A LoRA adapter for the assess step is the natural next step once more expert data exists.

Apertus Interview Coach — architecture

The code owns the conversation, the LLM fills slots · ~1.2 LLM calls per candidate answer · Apertus 1.5 8B, on-premise

